

FORET MONTMORENCY, QC, Canada

WMO: 712120

Lat: 47.3228N

Lon: 71.1483W

Elev: 673

StdP: 93.50

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-28.1	-34.3	0.2	-30.7	-31.4	0.2	-27.9	7.9	-11.9	7.1	-13.6	1.5	310	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	25.0	17.3	23.4	16.6	22.0	16.1	19.5	22.4	18.5	21.2	17.5	20.3	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.5	21.0	17.5	13.6	19.9	16.5	12.7	18.9	58.8	22.4	55.3	21.1	52.1	20.3	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.7	5.1	DB	-35.3	27.9	2.9	1.4	-37.4	28.9	-39.1	29.7	-40.7	30.6	-42.8	31.6	(4)
(5)			WB	-35.5	21.2	2.9	0.9	-37.5	21.9	-39.2	22.4	-40.8	22.9	-42.9	23.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-14.8	-13.5	-8.4	-0.3	6.8	12.0	15.0	13.7	9.9	3.7	-3.6	-10.8	(6)
(7)		DBStd	11.61	7.46	6.54	6.60	4.84	3.83	3.51	2.90	3.02	3.98	4.38	5.42	6.50	(7)
(8)		HDD10.0	3742	767	657	569	308	114	18	1	5	51	200	408	644	(8)
(9)		HDD18.3	6372	1026	890	828	558	357	191	109	144	254	454	658	902	(9)
(10)		CDD10.0	423	0	0	0	1	15	79	156	121	47	4	0	0	(10)
(11)		CDD18.3	11	0	0	0	0	0	2	6	2	1	0	0	0	(11)
(12)		CDH23.3	144	0	0	0	0	11	31	69	24	8	0	0	0	(12)
(13)		CDH26.7	9	0	0	0	0	1	1	7	1	1	0	0	0	(13)
(14)	Wind	WSAvg	2.0	2.1	2.2	2.4	2.4	2.0	1.9	1.8	1.7	1.8	1.9	2.0	1.9	(14)
(15)	Precipitation	PrecAvg	1415	123	91	95	101	101	120	150	125	130	128	132	126	(15)
(16)		PrecMax	1635	199	187	151	206	179	201	269	226	264	222	209	186	(16)
(17)		PrecMin	1171	38	46	29	40	42	55	30	59	68	42	40	58	(17)
(18)		PrecStd	145	40	38	31	46	35	41	53	40	40	48	41	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	3.7	11.5	16.7	25.0	26.1	27.6	25.8	24.3	19.0	12.1	4.2	(19)
(20)			MCWB	5.7	2.4	6.2	9.2	14.8	16.8	18.9	18.2	17.9	13.5	8.9	3.8	(20)
(21)		2%	DB	1.9	0.6	6.0	13.4	21.0	24.1	25.2	23.7	21.6	15.9	8.2	2.0	(21)
(22)			MCWB	1.7	-0.5	1.5	7.6	13.2	16.1	17.8	17.6	16.8	11.6	6.2	1.7	(22)
(23)		5%	DB	-1.0	-1.1	3.5	10.4	18.0	22.2	23.5	22.0	19.7	13.5	5.7	0.4	(23)
(24)			MCWB	-1.5	-2.1	0.8	5.6	11.7	15.2	16.9	16.5	15.9	10.7	4.5	-0.1	(24)
(25)		10%	DB	-3.5	-2.7	1.3	7.6	15.6	20.1	22.0	20.6	17.7	11.1	3.5	-1.4	(25)
(26)			MCWB	-4.1	-3.6	-0.4	3.6	10.4	14.2	16.5	16.1	14.7	9.1	2.6	-2.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	2.7	7.2	11.8	17.6	20.1	21.1	20.4	19.5	15.0	10.4	3.9	(27)
(28)			MCDWB	5.9	2.8	10.0	15.7	21.2	23.2	24.1	23.3	22.1	16.0	11.3	4.0	(28)
(29)		2%	WB	1.7	-0.4	2.9	8.8	15.2	18.2	19.7	18.9	18.1	13.0	6.9	1.6	(29)
(30)			MCDWB	1.9	0.4	4.6	11.8	18.8	21.1	22.6	21.4	20.4	15.5	7.7	1.9	(30)
(31)		5%	WB	-1.6	-2.1	1.0	5.9	13.0	16.8	18.7	17.9	16.7	11.0	4.7	-0.1	(31)
(32)			MCDWB	-1.2	-1.3	2.7	9.2	15.8	19.5	21.5	20.7	18.8	12.6	5.4	0.2	(32)
(33)		10%	WB	-4.1	-3.7	-0.3	4.2	11.2	15.5	17.6	16.9	15.3	9.3	2.6	-2.2	(33)
(34)			MCDWB	-3.7	-2.9	1.6	7.2	14.7	18.7	20.6	19.5	17.4	10.9	3.6	-1.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	11.8	12.3	10.8	12.5	12.4	11.9	11.6	8.7	8.3	9.9	(35)	
(36)			MCDBR	10.6	11.3	14.9	16.2	18.3	16.4	14.8	14.3	14.5	13.9	11.1	9.4	(36)
(37)			MCWBR	10.8	10.0	11.4	10.6	11.0	9.1	8.4	8.6	9.7	10.0	9.1	9.9	(37)
(38)		5% WB	MCDBR	10.7	10.3	11.7	13.7	13.9	11.5	11.3	11.0	11.9	11.4	9.7	9.5	(38)
(39)			MCWBR	10.7	9.5	9.7	10.1	10.0	7.7	7.4	7.4	8.9	9.3	8.9	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.244	0.259	0.293	0.321	0.333	0.356	0.371	0.353	0.336	0.311	0.285	0.258	(40)	
(41)		taud	2.338	2.288	2.247	2.356	2.468	2.406	2.371	2.427	2.474	2.534	2.493	2.356	(41)	
(42)		Ebn at Noon	860	918	930	934	931	906	886	890	877	851	806	792	(42)	
(43)		Edh at Noon	70	92	114	114	106	114	116	105	91	74	62	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.38	2.29	3.45	4.44	5.06	5.44	5.28	4.59	3.58	2.10	1.21	0.93	(44)	
(45)		RadStd	0.18	0.28	0.28	0.37	0.53	0.36	0.55	0.30	0.28	0.18	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page